TACTILE BUTTON SWITCH EXERCISE

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Use the Simulink Support Package for Arduino Hardware to develop and deploy an algorithm for a simple circuit containing an LED and a tactile button switch.	☐
2	Configure and use an Arduino Uno digital (PWM) output pin to control an LED in combination with a tactile push-button switch.	☐

2.0 Hardware Set-Up

The exercise involves connecting an LED in series with a tactile button switch, so that pressing the button controls whether the LED is on or off, as shown in Figure 1.

Note: While the ACE-Box can be used for this exercise, it is not required. Only the individual components listed below are needed.

Required hardware

- ATmega328P microcontroller (Arduino Uno)
- USB Type-B cable
- Breadboard
- LED
- 220 Ω resistor
- 2 × male-to-male breadboard wires
- Tactile button switch

Hardware Assembly Steps

1. Insert the tactile button switch so that its legs sit across the centre gap of the breadboard, with each pair of legs on opposite sides where no internal connection exists.
2. Use a male-to-male jumper wire to connect Pin 9 on the Arduino to one leg of the tactile switch (choose a column that connects to only that leg).

3. Insert a 220 Ω resistor so that one end is in the same column as the opposite leg of the tactile switch (the leg that is not connected to Pin 9). Place the other end of the resistor in a separate row.
4. Connect the long leg (anode) of the LED to the free end of the resistor. Complete the circuit by connecting the LED's short leg (cathode) to a GND pin on the Arduino using a male-to-male jumper wire.

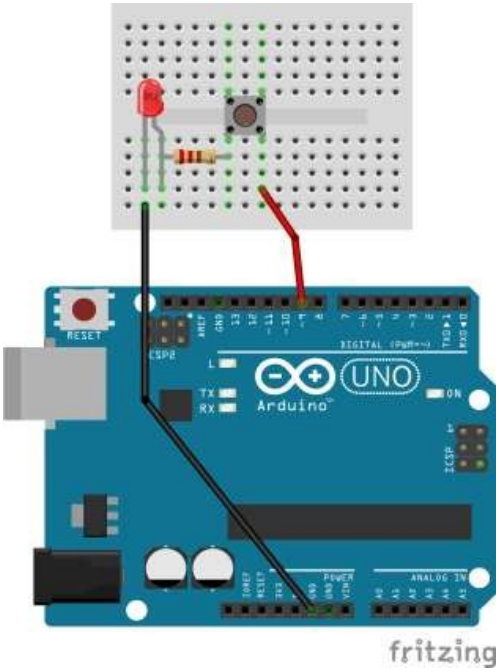


Figure 1: *Hardware Set-Up for the Tactile Button Switch Exercise*

3.0 Simulink Set-Up and Tactile Button Switch Testing

The tactile button switch adds no inputs or outputs to the Arduino, so the Simulink model is the same as in the LED (PWM) and On/Off Power Switch Exercises, as shown in Figure 2. For completeness, the full set-up is given again.

Simulink Model Construction Steps

1. From Simulink → Sources, add a Sine Wave block and connect it to an Abs (absolute value) block from Math Operations.
2. From Math Operations, connect a Slider Gain block. The lower (Low) and upper (High) limits of the gain constrain the output. For example, to limit the output to 2.3 V: 256 × 2.3 = 117.76 ≈ 117, so High would be set to 117. For this exercise, set High to 255 (equivalent to 5 V on the Arduino Uno).
3. From Simulink Support Package for Arduino Hardware → Common, add a PWM block, set the Pin number to 9, and connect three Scope blocks (from Sinks) to monitor the signals.
4. Deploy the model to the Arduino Uno and confirm that the LED brightness varies. Explore the effect of the gain value in the Slider Gain block.

Key Block Parameters to Modify

- Sine Wave (Sine type) → Sample based
- Sine Wave (Samples per period) → 300
- Sine Wave (Sample time) → 0.01 s

- Slider Gain (High) → 255

All other parameters may remain at their default values.

To recap how to run Simulink code generation and deploy the model to the Arduino Uno, see Section 4.0 of the LED (Digital) Exercise or visit Code Generation.

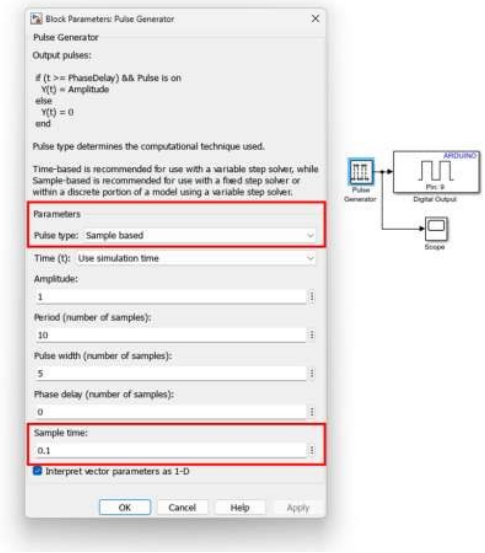


Figure 2: *Simulink Set-Up for the Tactile Button Switch Exercise*

With the model running, press and hold the tactile button switch and confirm that the LED only lights while the button is pressed.

4.0 Concluding Remarks

This exercise has shown how a tactile push-button switch can be combined with a PWM-controlled LED, and how a Simulink model is deployed to the ATmega328P microcontroller using the Simulink Support Package for Arduino Hardware. Unlike the latching on/off power switch, the tactile switch only completes the circuit while it is pressed.

Before completing the exercise, return to the Learning Outcomes in Table 1 and confirm that you are confident you can demonstrate each one. If so, tick the corresponding Achieved box.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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